Classic cryptosystems

Vigenère Tableau Cipher

- A set of substitution rules consists of the 26 Caesar ciphers, with shifts of 0 through 25.
- Given a key letter *x* and a plaintext letter y, the ciphertext letter is at the intersection of the row labeled *x* and the column labeled y; in this case the ciphertext is V.

Vigenère Tableau Cipher

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère Tableau Cipher: Example Encryption

- To encrypt a message, a key is needed that is as long as the message. Usually, the key is a repeating keyword.
- For example, if the keyword is *deceptive*, the message "we are discovered save yourself" is encrypted as follows:
- key: *deceptivedeceptivedeceptive*
- plaintext: wearediscoveredsaveyourself
- ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Vigenère Tableau Cipher: Example Decryption

- Decryption is equally simple.
- The key letter again identifies the row.
- The position of the ciphertext letter in that row determines the column, and the plaintext letter is at the top of that column.

Affine cipher

- The affine cipher is a type of substitution cipher, where each letter in an alphabet is mapped to its numeric equivalent, encrypted using a simple mathematical function, and converted back to a letter.
- Each letter is enciphered with the function
- $C = (ax + b) \bmod m$
- where modulus m is the size of the alphabet and a and b are the keys of the cipher.

Affine cipher Decryption

- The decryption function is

  $-x = a^{-1}(c - b) mod\ m$

- where a^{-1} is the modular multiplicative inverse of a modulo m. I.e., it satisfies the equation

  $-aa^{-1} mod\ m = 1$

Affine cipher: Example Encryption

- a should be chosen to be relatively prime to m (i.e. a should have no factors in common with m).
- Only the value of a has a restriction. The possible values that a could be are 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, and 25.
- There are 12 × 26 or 312 possible keys:
  - This number comes from the fact there are 12 numbers that are prime with 26 that are less than 26 (these are the possible values of *a*). Each value of *a* can have 26 different addition shifts (the *b* value)
- The plaintext to be encrypted is "AFFINE CIPHER", taking *a* to be 5, *b* to be 8, and *m* to be 26 since there are 26 characters in the alphabet being used.
- Thus, the encryption function for this example will be
  - $C = (5x + 8) \bmod 26$.

Affine cipher: Example Encryption

plaintext	A	F	F	I	N	E	C	I	P	H	E	R
x	0	5	5	8	13	4	2	8	15	7	4	17
$(5x + 8)$	8	33	33	48	73	28	18	48	83	43	28	93
$(5x + 8) \bmod 26$	8	7	7	22	21	2	18	22	5	17	2	15
ciphertext	I	H	H	W	V	C	S	W	F	R	C	P

Affine cipher: Example Decryption

- The decryption function is
  - $x = 21(y - 8) \bmod 26$
- where a^{-1} is calculated to be 21, and b is 8.
- $5(a^{-1}) mod\ 26 = 1$

ciphertext	I	H	H	W	V	C	S	W	F	R	C	P
y	8	7	7	22	21	2	18	22	5	17	2	15
$21(y-8)$	0	-21	-21	294	273	-126	210	294	-63	189	-126	147
$21(y-8) \bmod 26$	0	5	5	8	13	4	2	8	15	7	4	17
plaintext	A	F	F	I	N	E	C	I	P	H	E	R

Affine cipher: Example 2 Encryption

- Encrypt the plaintext “sail” using an affine cipher with encryption key (3,7)
- $s \rightarrow 18 \rightarrow 3 \cdot 18 + 7 \equiv 9 \pmod{26} \rightarrow J$
- $a \rightarrow 0 \rightarrow 3 \cdot 0 + 7 \equiv 7 \pmod{26} \rightarrow H$
- $i \rightarrow 8 \rightarrow 3 \cdot 8 + 7 \equiv 5 \pmod{26} \rightarrow F$
- $l \rightarrow 11 \rightarrow 3 \cdot 11 + 7 \equiv 14 \pmod{26} \rightarrow O$

One time pad

- The Vernam cipher, or one-time pad, is a provably secure cryptosystem
- Using a one-time pad to convert the letters to the bit string.
- The mapping between letters and bits is not secret.

One-Time Pad: Encryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Encryption: Plaintext ⊕ Key = Ciphertext

Plaintext:	h	e	i	l	h	i	t	l	e	r
	001	000	010	100	001	010	111	100	000	101
Key:	111	101	110	101	111	100	000	101	110	000
Ciphertext:	110	101	100	001	110	110	111	001	110	101
	s	r	l	h	s	s	t	h	s	r

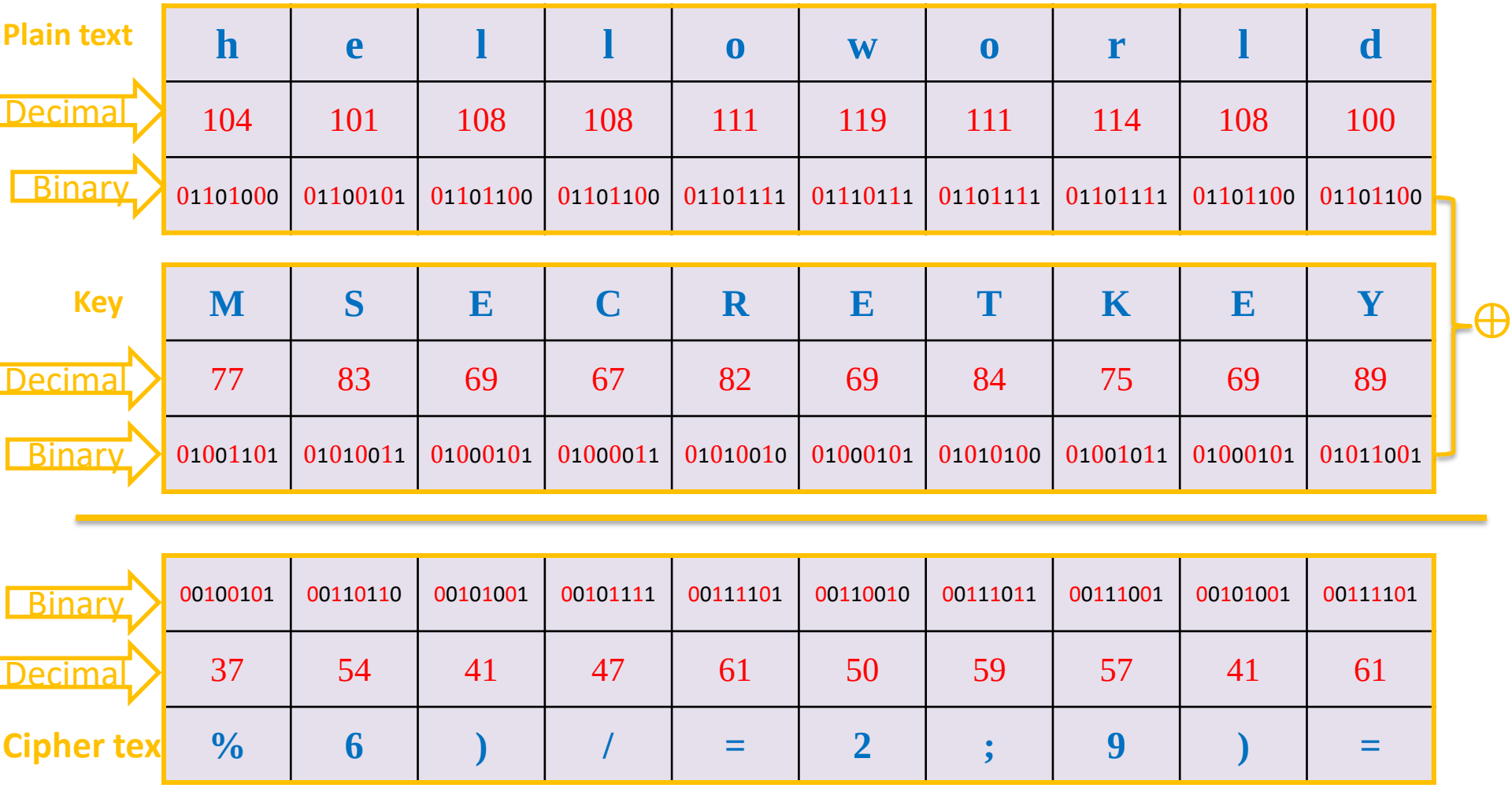
a	b	a XOR b
0	0	0
0	1	1
1	0	1
1	1	0

One-Time Pad: Decryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Decryption: Ciphertext ⊕ Key = Plaintext

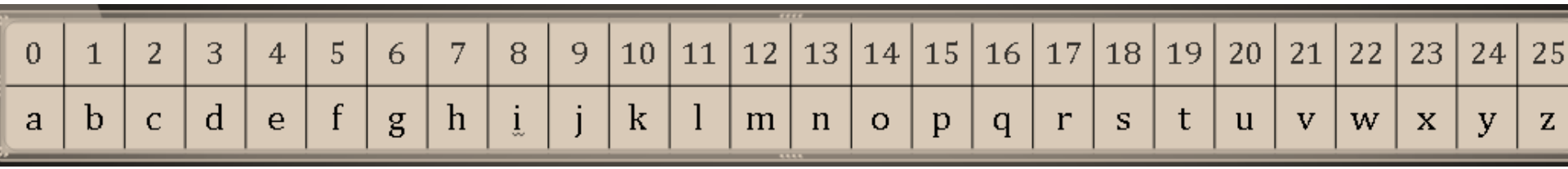
	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
Key:	111	101	110	101	111	100	000	101	110	000
Plaintext:	001	000	010	100	001	010	111	100	000	101
	h	e	i	l	h	i	t	l	e	r


Plain text
h e l l o w o r l d
Decimal
104 101 108 108 111 119 111 114 108 100
Binary
01101000 01100101 01101100 01101100 01101111 01110111 01101111 01101111 01101100 01101100
Key
M S E C R E T K E Y
Decimal
77 83 69 67 82 69 84 75 69 89
Binary
01001101 01010011 01000101 01000011 01010010 01000101 01010100 01001011 01000101 01011001
⊕
Binary
00100101 00110110 00101001 00101111 00111101 00110010 00111011 00111001 00101001 00111101
Decimal
37 54 41 47 61 50 59 57 41 61
Cipher tex
% 6) / = 2 ; 9) =

One-Time Pad 2

- This technique is to combine the key and the message using modular addition
- The numerical values of corresponding message and key letters are added together, modulo 26.
- So, if key is **XMCKL** and the message is **hello**, then the coding would be done as follows:

One-Time Pad 2 (Encryption)



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z

```
      h         e         l         l         o  message
    7 (h)     4 (e)    11 (l)    11 (l)    14 (o) message
+  23 (X)    12 (M)     2 (C)    10 (K)    11 (L) key
=  30        16        13        21        25     message + key
=   4 (E)    16 (Q)    13 (N)    21 (V)    25 (Z) (message + key) mod 26
       E         Q         N         V         Z  → ciphertext
```

One-Time Pad 2 (Decryption)

```
      E        Q        N        V        Z  ciphertext
    4 (E)   16 (Q)   13 (N)   21 (V)   25 (Z) ciphertext
-  23 (X)   12 (M)    2 (C)   10 (K)   11 (L) key
= -19        4       11       11       14     ciphertext - key
=   7 (h)    4 (e)   11 (l)   11 (l)   14 (o) ciphertext - key (mod 26)
      h        e        l        l        o  → message
```

Drawbacks of OTP

- The key must be the same size as the message being sent.
- The key must be truly random.
- Keys must never be reused.
- Keys must be securely shared between the sending and receiving parties.

Cryptanalysis

- Suppose the attacker catches Alice's ciphertext: EQNVZ.
- If attacker tried every possible key, she would find that the key XMCKL would produce the plaintext hello
- But she would also find that the key TQURI would produce the plaintext later.
- If the key is **not truly random**, it is possible to use statistical analysis to determine which of the plausible keys is more likely to be the correct one.
- If a key is **reused**, it will noticeably be the only key that produces sensible plaintexts from both ciphertexts